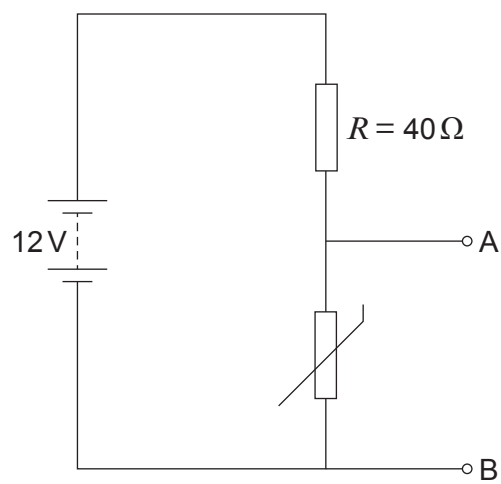


7. The circuit below is used to investigate how the resistance of a thermistor changes as temperature increases.

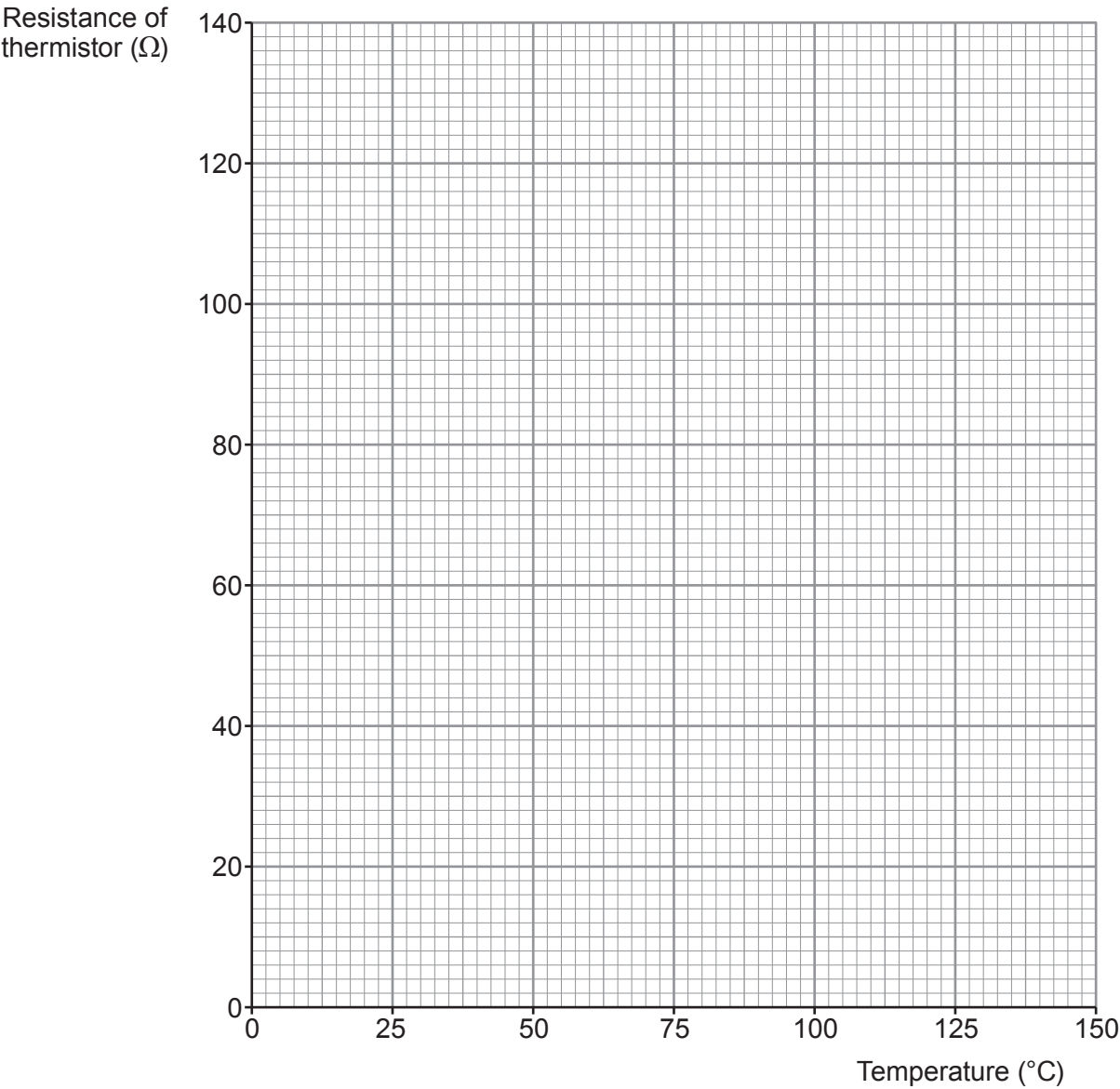


- (a) **Add an ammeter and a voltmeter** to the diagram so the necessary measurements can be taken. [2]
- (b) **Plot the results shown** below on the grid opposite and draw a curve of best fit. [2]

Temperature (°C)	Resistance of thermistor (Ω)
0	120
25	92
50	70
75	52
100	40
125	30



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- (c) The resistance of the resistor, R , remains constant at $40\,\Omega$.
- (i) Calculate the voltage across the terminals AB when the temperature is $100\,^{\circ}\text{C}$. [3]

voltage = V



- (ii) Tom says as temperature increases the current increases.
Explain whether you agree.

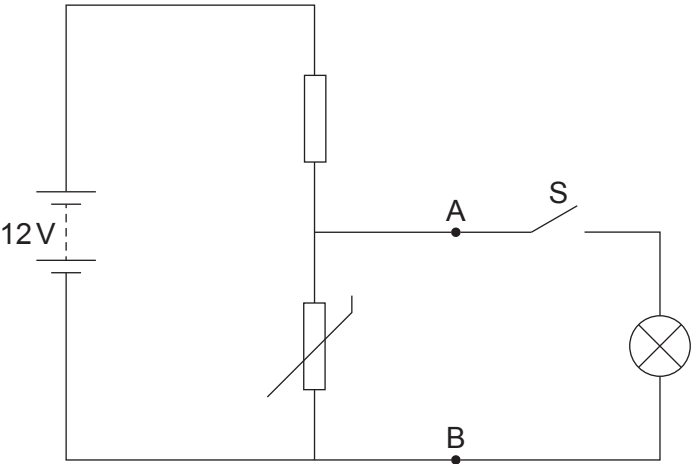
[2]

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- (d) A lamp and a switch are now connected across the terminals AB, as shown in the diagram below.



- (i) The lamp has a power of 3W at 12V.
Use equations from page 2 to calculate the resistance of the lamp at this power and voltage.

[3]

resistance = Ω



(ii) The switch, S, is now closed. Explain, without calculation, what happens to the total resistance of the circuit. [2]

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14

